Sahaj Bhadja

(213) 284-4651 | bhadja@usc.edu | LinkedIn | GitHub | Portfolio

Summary

Full-stack software engineer with 2+ years building customer-support workflows, secure interfaces, APIs, and reliable financial applications. Experienced in React, TypeScript, Python, Flask, REST APIs, SQL, access controls, integrations, automated testing, observability, and production support.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Analysis of Algorithms, Database Systems, Web Technologies, Advanced NLP

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – Present*
- Built Python APIs, Azure SQL services, and agent-style dashboards for 500+ users, surfacing contextual data through secure interfaces while reducing response latency from 1.2 seconds to 350 milliseconds.
- Implemented reliable stateful workflows for 5,000+ monthly jobs using explicit states, retries, reconciliation, and failure isolation, tripling throughput while improving correctness, observability, maintainability, and support across production systems.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Developed React interfaces and Flask REST services handling 10,000+ daily requests, implementing user context, status tracking, secure actions, and history workflows while resolving production issues for 150+ users.
- Created concurrent processing for 1,000+ files with automated validation, AWS RDS persistence, structured logging, and error handling, reducing resolution turnaround 60% while maintaining reliable application behavior under production load.

Aasma Technology Solutions **India**
Software Developer Engineer – SecureSync *August 2023 – July 2024*
- Developed Python services, SQL integrations, and ETL workflows using AWS Glue, S3, and REST APIs, improving response times 30% and data-processing efficiency 68% across customer-facing applications in production.
- Maintained 20+ automated unit and integration suites with Jenkins, PyTest, and Mockito across three platforms, accelerating defect diagnosis, code review feedback, and confidence in reliable production releases.

Aasma Technology Solutions **India**
Full Stack Developer Intern – VibrantChem *May 2023 – August 2023*
- Developed modular React and Node.js features across eight authentication, permissions, notifications, and support workflows, synchronizing 2 million+ Azure SQL records through secure, maintainable services and real-time WebSockets.
- Automated Docker builds, tests, staged deployments, and production monitoring through Jenkins and Azure, reducing release time 50% while improving access-control reliability, deployment consistency, and operational visibility across environments.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Built Flask APIs and responsive workflows for reservations, status updates, support actions, and payments across 50,000+ records, translating operational requirements into maintainable features with secure database access.
- Implemented Firebase persistence, webhooks, and payment synchronization with structured validation and logging, reducing manual payment handling 70% while improving issue visibility, transaction reliability, and responsive user support.

Skills

Languages & Frontend: TypeScript, JavaScript, Python, Java, SQL, React, Next.js, HTML5, CSS3, Tailwind CSS, responsive interfaces, component design
Backend & Tooling: Flask, FastAPI, Node.js, REST APIs, microservices, WebSockets, business logic, workflow automation, case workflows, API integration, access controls
Data & Cloud: PostgreSQL, Azure SQL, SQL Server, MySQL, MongoDB, Firebase, AWS RDS, AWS, Azure, S3, Blob Storage, data modeling, query optimization
Quality & Operations: Git, Docker, Kubernetes, Jenkins, CI/CD, PyTest, Mockito, unit testing, integration testing, code review, debugging, logging, monitoring, observability, production support, Agile

Projects

FinNexus | Live | *TypeScript, React, REST APIs, WebSockets, MongoDB, Kubernetes*
- Built secure financial workflows supporting 50,000+ users and one million transactions, integrating transaction context, REST APIs, real-time synchronization, granular filters, MongoDB persistence, monitoring, and automated Kubernetes deployment.

HMIS | Live | *TypeScript, React, REST APIs, Appwrite, Sentry, Twilio*
- Built case-management workflows for appointments, records, and consent-based access using React, REST APIs, authentication, notifications, and Sentry monitoring, supporting 100+ daily interactions while reducing administrative workload 30%.

Big-Brain | Live | *LLaMA-2, FAISS, PostgreSQL, REST APIs, Clerk, Convex*
- Built an agent-style document workspace using LLaMA-2, FAISS, and REST APIs, surfacing contextual information through secure access controls, conversational search, and real-time collaboration across 50,000+ user documents.